

Sol. IV

POKHARA UNIVERSITY

Level: Bachelor
 Programme: BE
 Course: Analysis and Design of Algorithm (New)

Semester: Spring

Year : 2024
 Full Marks: 100
 Pass Marks: 45
 Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the measures that we use to measure the efficiency of an algorithm? Write an algorithm to find an element from an array and discuss its best and worst case while finding an element. 7
- b) Explain the master's method for solving the recurrence relations. Solve the following recurrence relations using the master's method. 8
 $T(n) = T(2n/3) + 1$
 $T(n) = 3T(n/2) + n^2$
 $T(n) = 2T(n/2) + n \log(n)$
2. a) Provide detailed algorithms or pseudocode for the search operations in a binary search tree. Analyze the time and space complexity of the operation. 7
- b) Write an algorithm for quick-sort and trace out the algorithm for the given array $A[] = \{-4, 1, 25, 50, 8, 10, 23\}$. Also write the computational complexity. 8

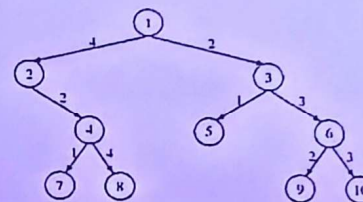
OR

Describe Strassen's Matrix multiplication and its role in reducing the time complexity of matrix multiplication.

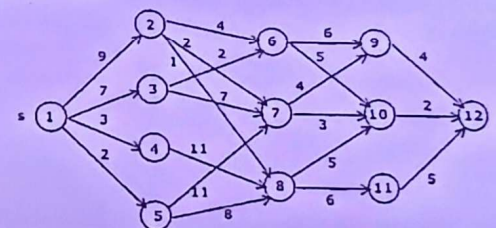
3. a) Consider the following tasks with their deadlines and profits. Schedule the tasks in such a way that they produce maximum profit after being executed using greedy method: 7

S. No.	1	2	3	4	5
Jobs	J1	J2	J3	J4	J5
Deadlines	2	2	1	3	4
Profits	20	60	40	100	80

- b) What is the real-life application of tree vertex splitting problem? For $\delta = 5$, add a booster to the following tree. 8



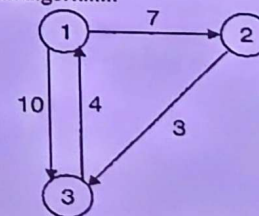
4. a) Use the dynamic programming approach to solve the given 7 multistage graph. 7



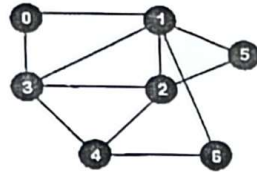
- b) Using dynamic programming, devise an algorithm to edit the string "sitting" into "kitten" with the minimum number of operations. Illustrate the steps and transformations necessary to achieve this conversion. 8

OR

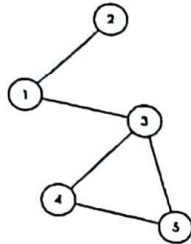
Find the shortest path between every pair of vertices in a given graph using floyd-Warshall algorithm.



5. a) Write differences between DFS and BFS. Generate the spanning tree of the following graph using DFS. 8



- b) What is biconnected components? Determine biconnected components of a given graph using DFS. 7



6. a) How can the sum of subset problem be approached using backtracking for a given set {3, 34, 4, 12, 5, 2} and a target sum of 9? 7
 b) What do you mean by graph coloring problem? How this problem could be solved with the help of Backtracking algorithm, explain with example. 8
7. Write short notes on: (Any two) 2x
 a) Flow Shop Scheduling
 b) Fractional knapsack problem
 c) Finding minimum and maximum

ADA(New) 2024(JP)

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Attempt all the questions.

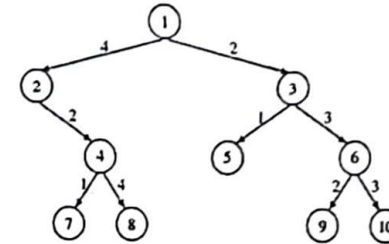
1. a) Why is asymptotic of algorithm analysis important? Discuss the concepts of best-case, worst-case, and average-case complexities with a suitable example. Additionally, illustrate the example using a graph to explain the differences. 8
- b) What is a randomized algorithm? Write differences between Las Vegas and Monte Carlo algorithms with examples. 7
2. a) Describe Strassen's Matrix multiplication and its role in reducing the time complexity of matrix multiplication. 8
- b) Consider the following tasks with their deadlines and profits. Schedule the tasks in such a way that they produce maximum profit after being executed using greedy method 7

S. No.	1	2	3	4	5
Jobs	J1	J2	J3	J4	J5
Deadlines	3	2	1	3	4
Profits	20	10	40	50	70

OR

Generate a single file using the optimal merge pattern for the given files (f1, f2, f3, f4, f5) with their respective number of elements (25, 30, 15, 50, 20).

3. a) What are the real-life applications of the tree vertex splitting problem? Given a tree with degree $\delta=5$, demonstrate how to add a booster to the tree and explain the process. 8



- b) Using dynamic programming, devise an algorithm to convert the string "money" into "monkey" with the minimum number of operations. Illustrate the steps and transformations necessary to achieve this conversion. 7
4. a) What is the optimal binary search tree? Using dynamic approach, build optimal binary tree for the given values. 7

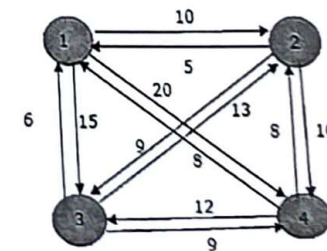
Keys	10	20	30	40
Frequency	4	2	6	3

OR

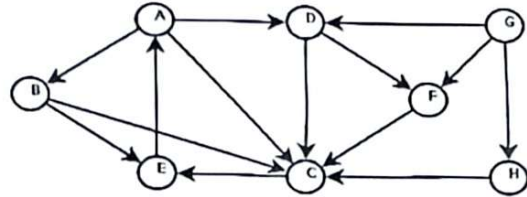
Two jobs have to be scheduled on three processors. Generate a non-preemptive schedule. The following matrix gives the task times:

2	0
3	3
5	2

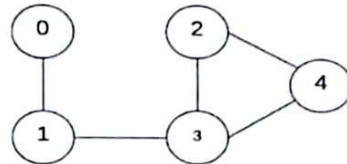
- b) Perform travelling salesman problem using dynamic programming in given graph. 8



5. a) Write differences between DFS and BFS. Generate the spanning tree of the following graph using DFS. 8



- b) What is biconnected components? Determine biconnected components of a given graph using DFS. 7



6. a) What is the backtracking approach in problem-solving, and why is it useful? Explain the Sum of Subset problem and write a backtracking algorithm to find all subsets of a given set whose sum equals a target value. Illustrate the algorithm with an example. 8
- b) What do you mean by graph coloring problem. How this problem could be solved with the help of Backtracking algorithm, explain with example. 7
7. Write short notes on: (Any two) 2×5
- Binary search tree
 - Priority queues
 - Fractional Knapsack and 0/1 Knapsack

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year : 2021
Programme: BE		Full Marks: 100
Course: Analysis and Design of Algorithm		Pass Marks: 45
		Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.*

1. a) Why do you need the algorithm analysis? Explain the best, worst and average case complexities with suitable example. 8
- b) Explain the master method for solving the recurrence relations. Solve the following recurrence relations using this method. 7
 - i. $T(n) = 3T(n/2) + n$
 - ii. $T(n) = 2T(n/4) + \sqrt{n}$
2. a) Explain Binary Search Tree (BST)? Write an algorithm to insert an element to the binary search tree and find the time complexity of your algorithm. 7
- b) Briefly explain the Queue data structure. Write an algorithm to add and remove an element from the circular queue and compute the complexity of your algorithm. 8
3. a) Consider five items along with their respective weights and profit values 7

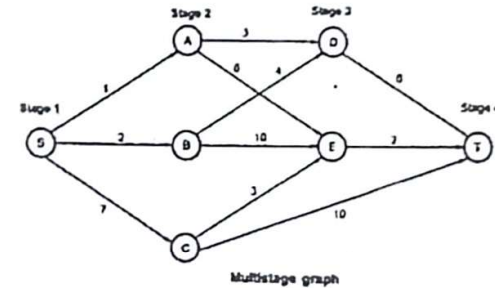
Items I = <11, 12, 13, 14, 15>

Weights w=<5, 10, 20, 30, 40>

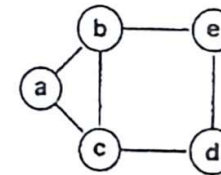
profit value v=<30, 20, 100, 90, 160>

The Knapsack has capacity W=60. Find an optimal solution to the Knapsack Problem
- b) Explain how you use the divide and conquer approach to solve Strassen's Matrix Multiplication. 8
4. a) Write an algorithm for quick-sort and trace out the algorithm for the following array A[] = { 16,7,15,14,18,25,55,32 }. 8

- b) Briefly explain the dynamic programming method for problem solving. What is the basic difference between Dynamic programming and Greedy method? 7
5. a) Find the shortest path in the following multistage graph using dynamic programming from vertex S to T. 8



- b) Define spanning tree. Generate spanning tree for graph below using both BFS and DFS method. 7



6. a) Define articulation point and bi-connected component of the graph? Write the algorithm for finding articulation point in the graph with suitable example. 7
- b) Explain about the complexity classes P, NP and NP complete with suitable examples. 8
7. Write short notes on: (Any two) 2x5
 - a) Travelling Salesman Problem
 - b) Optimal Storage on Tapes Problem
 - c) Graph Colouring Problem

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2022
 Programme: BE Full Marks: 100
 Course: Analysis and Design of Algorithm Pass Marks: 45
 Time : 3hrs.

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Attempt all the questions.

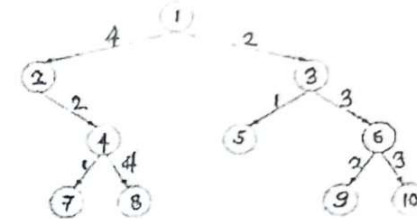
1. a) Define algorithm. Why is analysis of algorithms necessary? Explain asymptotic notations BigO, Big Θ and Big Ω . 8
- b) Define Binary Search Tree. Write an algorithm to search an element in the binary search tree and find the time and space complexity of your algorithm. 7
2. a) What is the difference between fractional knapsack and 0/1 knapsack. Consider the following 7 items and the knapsack has capacity of 15. Find an optimal solution using fractional knapsack. 8

Objects	1	2	3	4	5	6	7
Profits	10	5	15	7	6	18	3
Weights	2	3	5	7	1	4	1

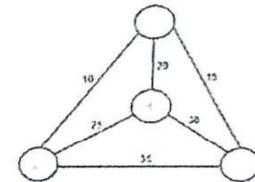
- b) Consider the set of given jobs as shown. Find a sequence of jobs, which will give maximum profit 7

Jobs	J1	J2	J3	J4	J5	J6
Deadlines	5	3	3	2	4	2
Profits	200	180	190	300	120	100

3. a) What is the real-life application of tree vertex splitting problem? For $\delta = 5$, add booster to the following tree. 8



- b) Explain and analyze the Floyd's Warshall algorithm for all pair shortest path problem. Trace the algorithm with suitable example. 8
4. a) Find the cost of travelling from vertex one to rest of the vertex for a sales man having minimum cost, travelling all the cities. 8



- b) What do you mean by an articulation point in a network? What is the effect of deleting it in communication networks? When we say that a graph is bi-connected, What does it mean? Explain with a neat diagram. 7
5. a) What is the backtracking method for problem solving? Explain how you solve the 8-queen problem using the backtracking method. 7
- b) Write an algorithm for finding minimum and maximum elements in array. Also find its complexity. 8
6. a) Write an Algorithm for push and pop operation of stack. Also analyze the complexity. 8
- b) Explain Multistage graph problem with suitable example. 7
7. Write short notes on: (Any two) 2x5
 - a) String Editing Problem
 - b) Sum of Subset Problem
 - c) BFS and DFS

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2023
 Programme: BE Full Marks: 100
 Course: Analysis and Design of Algorithm Pass Marks: 45
 Time : 3hrs.

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Attempt all the questions.

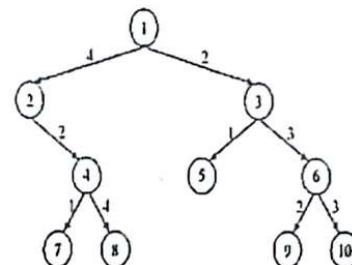
1. a) What is the need for algorithm analysis? How would you perceive the best, worst and average case complexities? Give an example, along with a suitable graph. 7
- b) Explain the master's method for solving the recurrence relations. Solve the following recurrence relations using the master's method. 8
 - i. $T(n) = 3T(n/2) + n$
 - ii. $T(n) = 2T(n/4) + \sqrt{n}$
2. a) Briefly explain the Queue data structure. Write an algorithm to add an element from the circular queue and compute the complexity of your algorithm. 7
- b) Briefly explain the binary search tree. Write an algorithm to remove an element from the binary search tree and compute the complexity of your algorithm. 8
3. a) What is the difference between fractional knapsack and 0/1 knapsack. Consider the following 7 items and the knapsack has a capacity of 15. Find an optimal solution using fractional knapsack. 7

Object(O)	1	2	3	4	5	6	7
Profits(P)	10	5	15	7	6	18	3
Weights(w)	2	3	5	7	1	4	1

- b) Consider the set of given jobs as shown. Find a sequence of jobs, which will give maximum profit.

Jobs	J1	J2	J3	J4	J5	J6
Deadlines	5	3	3	2	4	2
Profits	200	180	190	300	120	100

4. a) What is the real-life application of tree vertex splitting problem? For $\delta = 5$, add a booster to the following tree. 7



- b) Explain the optimal merge pattern algorithm used for merging multiple sorted sequences. How does it minimize the total number of comparisons and improve efficiency? 8
5. a) Discuss the similarities and differences between divide and conquer, greedy method and dynamic programming. How do these approaches differ in their problem-solving strategies and efficiency? 7
- b) Explain "Principle of Optimality" in dynamic programming. Let $X = \text{moon}$ and $Y = \text{spoon}$. Find the minimum cost required to transform X into Y , if each insertion and deletion takes 1 unit and change takes 2 units of time. 8
6. a) Design a three-stage system with device types D1, D2 and D3. The costs are \$30, \$15, and \$20 respectively. The cost of the system is to be no more than \$105. The reliability of each device is 0.9, 0.8, 0.5 respectively. 8
- b) Define the articulation point and bi-connected component of the graph. Write an algorithm to determine biconnected components in a graph with a suitable example. 7
7. Write short notes on: (Any two) 2×5
 - a) Graph Coloring Problem
 - b) Sum of subset problem
 - c) Strassen's Matrix Multiplication

POKHARA UNIVERSITY

Level: Bachelor

Semester : Spring

Year : 2023

Programme: BE

Full Marks: 100

Course: Analysis and Design of Algorithm

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the measures that we use to measure the efficiency of an algorithm? Write an algorithm to find the largest number from an array of numbers and discuss its best and worst case. 7
- b) Briefly explain the binary search tree. Write an algorithm to add an element from the binary search tree and compute the complexity of your algorithm. 8
2. a) Briefly explain a circular queue. Write an algorithm to add and remove an element from the circular queue and compute the complexity of your algorithm. 7
- b) Consider the following tasks with their deadlines and profits. Schedule the tasks in such a way that they produce maximum profit after being executed – 8

S. No.	1	2	3	4	5
Jobs	J1	J2	J3	J4	J5
Deadlines	2	2	1	3	4
Profits	20	60	40	100	80

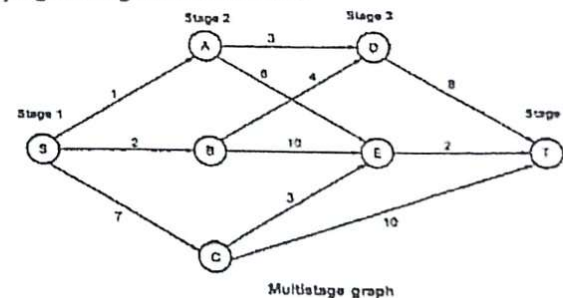
3. a) What is the difference between a fractional knapsack and a 0/1 knapsack? Consider the following 5 items and the knapsack has a capacity of 60. Find an optimal solution to the knapsack problem. 7

Objects	11	12	13	14	15
Profits	30	20	100	90	160
Weights	5	10	20	30	40

- b) Write an algorithm to find minimum and maximum elements among n elements and analyze its complexity. 8
- a) Compare divide and conquer, greedy method, dynamic programming and Backtracking approach of problem solving? Explain with examples. 7
- b) Explain "Principle of Optimality" in dynamic programming. Let X =sunday and Y =saturday. Find the minimum cost required to transform X into Y , if each insertion and deletion takes 1 unit and change takes 2 units of time. 8
- a) What is the optimal binary search tree? Using a dynamic approach, build an optimal binary tree for the given values. 8

Keys	10	20	30	40
Frequency	4	2	6	3

- b) Define the articulation point and bi-connected component of the graph. Write an algorithm to determine biconnected components in a graph with a suitable example. 7
- a) What do you mean by graph coloring problem? How this problem could be solved with the help of Backtracking algorithm, explain with algorithm. 8
- b) Find the shortest path in the following multistage graph using dynamic programming from vertex S to T . 7



Write short notes on: (Any two)

- a) Tree vertex Splitting
- b) BFS and DFS
- c) Hamiltonian Cycle

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Attempt all the questions.

1. a) What is the need for algorithm analysis? Explain the best, worst and average case complexities? Give an example, along with a suitable graph. 7
- b) What is a randomizer? Explain different types of randomized algorithms. 8
2. a) Briefly explain the Stack data structure. Write an algorithm to remove an element from the stack and compute the complexity of your algorithm. 7
- b) How are binary trees structured and utilized, and what are their key traversal methods? Also compute the complexity of your algorithm. 8
3. a) What is the difference between a fractional knapsack and a 0/1 knapsack? Given the following 7 items and a knapsack with a capacity of 15, find an optimal solution to the knapsack problem using the fractional knapsack method. 8

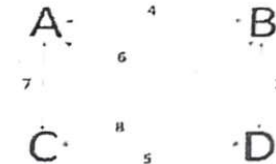
Objects	1	2	3	4	5	6	7
Profits	5	10	15	7	7	9	4
Weights	1	3	1	4	1	3	2

- b) Write an algorithm for merge sort and trace the algorithm for the given array $A[] = \{-10, 1, 23, 50, 4, 9, -4\}$. Also, write the computational complexity. 7
4. a) Write algorithm for job sequence with deadlines and solve the given problem. 7

Job	1	2	3	4	5
Deadline	2	1	2	1	3
Profit	100	50	75	80	90

Page 1 of 2

- b) Given a collection of cities and the distances between each pair of cities, using dynamic programming, solve the traveling salesman problem to find the shortest possible route that visits every city exactly once and returns to the starting point. 8



5. a) Explain "Principle of Optimality" in dynamic programming. Let X =room and Y =spoon. Find the minimum cost required to transform X into Y , if each insertion and deletion takes 1 unit and change takes 2 units of time. 8
- b) Discuss the concept of Minimum Spanning Tree (MST) and compare Prim's and Kruskal's algorithms for finding an MST. 7
6. a) How can the subset sum problem be solved using backtracking for $n=4$, weights $w = \{11, 13, 24, 7\}$, and a target sum $m=31$? 8
- b) Explain BFS and DFS as Graph Traversal Techniques with examples and analyze their performance in terms of time, space, completeness and optimality. 7
7. Write short notes on: (Any two) 2*5
 - a) Optimal Merge Pattern
 - b) Single source shortest path algorithm
 - c) Graph Coloring Problem

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